

Program 4

Project Structure

```
.  
├── Dockerfile  
├── index.js  
├── node_modules  
├── package.json  
└── package-lock.json
```

STEP 1: Create the Dockerfile and add the content:

```
1rv24mc048_Jayesh@Jayesh:~/Desktop/DEVOPS_Lab/lab2$ cat Dockerfile  
FROM node:20-alpine AS builder  
  
WORKDIR /app  
  
COPY package.json package-lock.json ./  
RUN npm install  
  
COPY . .  
RUN npm run build  
  
FROM node:20-alpine  
  
WORKDIR /app  
  
COPY --from=builder /app/package.json ./  
COPY --from=builder /app/package-lock.json ./  
COPY --from=builder /app/dist ./dist  
COPY --from=builder /app/node_modules ./node_modules  
  
EXPOSE 3000  
CMD ["node", "dist/index.js"]1rv24mc048_Jayesh@Jayesh:~/Desktop/DEVOPS_Lab/lab2$
```

STEP 2: Make a folder src and create a file → index.js d add the following content :

```
1rv24mc048_Jayesh@Jayesh:~/Desktop/DEVOPS_Lab/lab2$ cat src/index.js  
const express = require('express');  
const app = express();  
const port = 3000;  
  
app.get('/', (req, res) => {  
  res.send('Hello Deepika Maam');  
});  
  
app.listen(port, () => {  
  console.log(`Example app listening at http://localhost:${port}`);  
});  
1rv24mc048_Jayesh@Jayesh:~/Desktop/DEVOPS_Lab/lab2$ █
```

Step 3 : Create a File package.json

Initialize it with:

npm init -y (also creates the file package-lock.json)

```
1rv24mc048_Jayesh@Jayesh:~/Desktop/DEVOPS_Lab/lab2$ cat package.json
{
  "name": "lab2",
  "version": "1.0.0",
  "description": "",
  "main": "index.js",
  "scripts": {
    "start": "node dist/index.js",
    "build": "mkdir -p dist && cp src/* dist/",
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "dependencies": {
    "express": "^5.1.0"
  }
}
1rv24mc048_Jayesh@Jayesh:~/Desktop/DEVOPS_Lab/lab2$
```

STEP 4: Install node modules with the following command

npm install

STEP 5: Execute the Docker build command to build image

```
Terminal
1rv24mc048_Jayesh@Jayesh:~/Desktop/DEVOPS_Lab/lab2$ dockerscript
Enter the Name of the Image you want to create:
lab2
Image lab2 does not exist. Building it now...
[+] Building 8.3s (15/15) FINISHED                                docker:default
=> [internal] load build definition from Dockerfile                0.1s
=> => transferring dockerfile: 423B                                0.0s
=> [internal] load metadata for docker.io/library/node:20-alpine  3.2s
=> [internal] load .dockerignore                                  0.0s
=> => transferring context: 2B                                       0.0s
=> [internal] load build context                                  0.4s
=> => transferring context: 219.84kB                                0.4s
=> [builder 1/6] FROM docker.io/library/node:20-alpine@sha256:6178e78b972f79c335df281f4b7674a2d85071aa 0.0s
=> CACHED [builder 2/6] WORKDIR /app                               0.0s
=> CACHED [builder 3/6] COPY package.json package-lock.json ./    0.0s
=> CACHED [builder 4/6] RUN npm install                            0.0s
=> [builder 5/6] COPY . .                                          1.5s
=> [builder 6/6] RUN npm run build                                 2.3s
=> CACHED [stage-1 3/6] COPY --from=builder /app/package.json ./  0.0s
=> CACHED [stage-1 4/6] COPY --from=builder /app/package-lock.json ./ 0.0s
=> CACHED [stage-1 5/6] COPY --from=builder /app/dist ./dist      0.0s
=> CACHED [stage-1 6/6] COPY --from=builder /app/node_modules ./node_modules 0.0s
=> exporting to image                                              0.0s
=> => exporting layers                                              0.0s
=> => writing image sha256:8c05496aafa469d0eaa7b9004a4b28fed9a875cf19fd6ea470a75397862499c0 0.0s
=> => naming to docker.io/library/lab2                             0.0s
```

How to Generate a Personal Access Token (Classic)

Log in to your GitHub account.

Go to Settings → Developer settings → Personal access tokens → Tokens (classic)

Click Generate new token → Generate new token (classic)

Step 1: Initialize a Local Repository

git init

Rename the default branch to main (recommended):

git branch -M main

Step 2: Add and Commit Files

git add .

Commit them with a message:

git commit -m "Initial commit"

Step 3: Connect to a Remote GitHub Repository

git remote add origin github.com

Step 4: Using username and PAT

git push -u origin main

